

TITLE : RTL Design and Functional Simulation

Objective : Design an 8-bit ALU

RTL Code:

```
module alu(  
input [7:0] A,  
input [7:0] B,  
input [2:0] opcode,  
output reg [7:0] result  
);
```

```
always @(*) begin  
case(opcode)  
3'b000: result = A + B;  
3'b001: result = A - B;  
3'b010: result = A & B;  
3'b011: result = A | B;  
3'b100: result = A ^ B;  
default: result = 0;  
endcase  
end
```

```
endmodule
```

Testbench Code:

```
`timescale 1ns/1ps  
  
module alu_tb;  
  
reg [7:0] A, B;  
reg [2:0] opcode;  
wire [7:0] result;
```

```
alu uut (  
  .A(A),  
  .B(B),  
  .opcode(opcode),  
  .result(result)  
);
```

```
initial begin
```

```
  $dumpfile("dump.vcd");  
  $dumpvars(0, alu_tb);
```

```
  A = 10; B = 5; opcode = 0;  
  #10;
```

```
  A = 10; B = 5; opcode = 1;  
  #10;
```

```
  A = 10; B = 5; opcode = 2;  
  #10;
```

```
  A = 10; B = 5; opcode = 3;  
  #10;
```

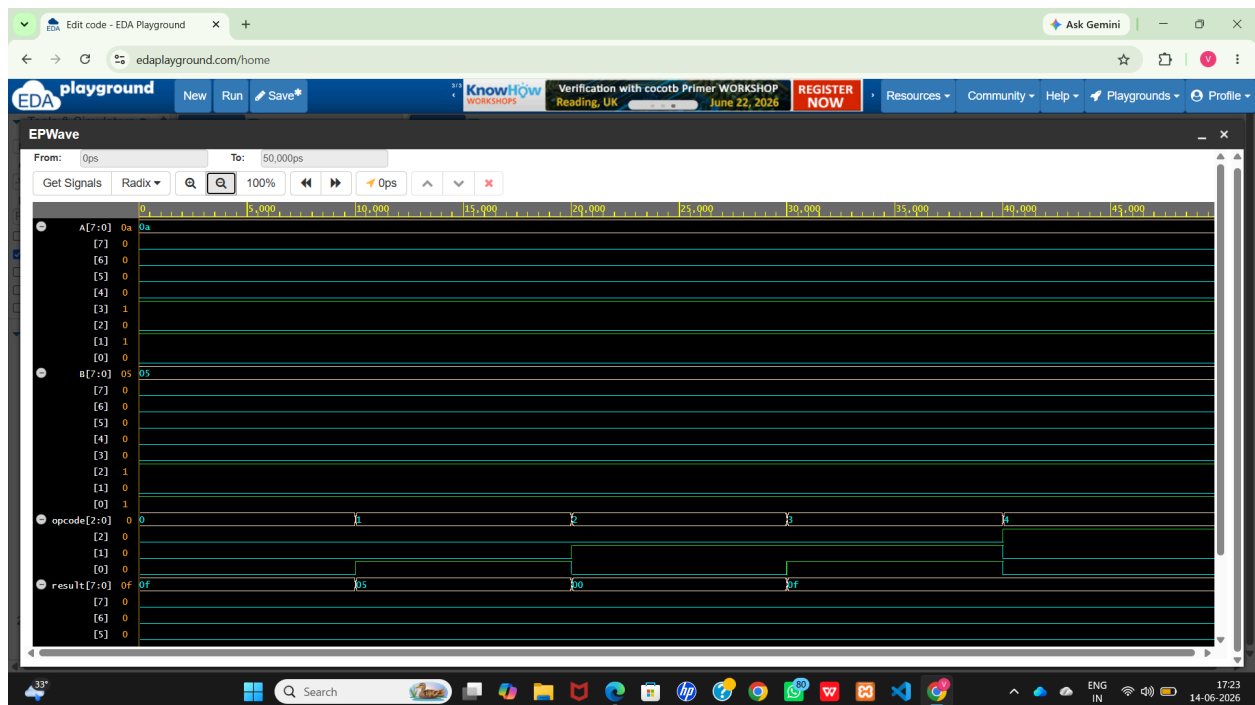
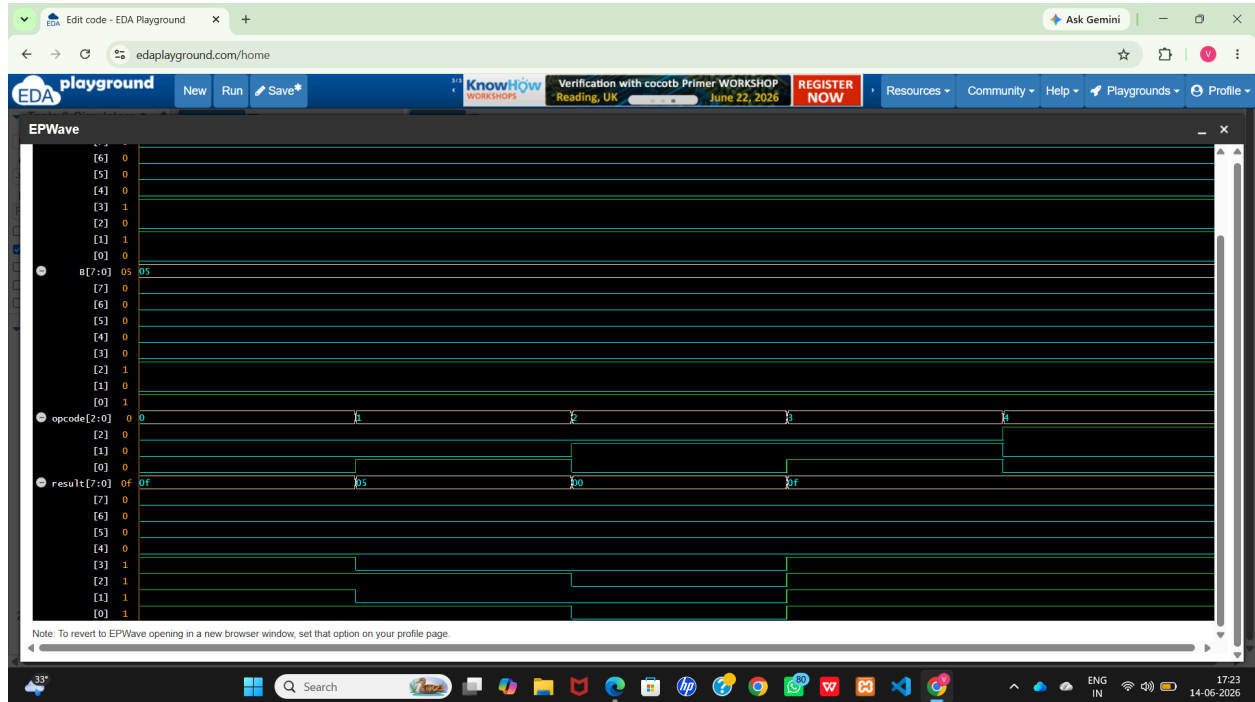
```
  A = 10; B = 5; opcode = 4;  
  #10;
```

```
  $finish;
```

```
end
```

```
endmodule
```

Simulation Result:



Conclusion :

The ALU was verified successfully.

